Southern New Hampshire University

# CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

**Tip:** You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

## System Components and Design (Completed)

### Purpose

*What is the purpose of this project? Who is the client and what do they want their system to be able to do?*

- The client for this project is Liam, the owner of Driver-Pass, and his IT officer Ian. Liam wants to create a system that provides better driving training through online courses.

### System Background

*What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?*

- He wants the system to be able to have practice tests for the users.
- The issue is that so many people fail their driving test at DMVs so what better way to prevent that than to practice online through accessible courses.
- The system should allow access to the data from anywhere (device).

### Objectives and Goals

*What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?*

- Users should be able to download important data to be used outside of the system.
- Liam wants to be able to access data online from any computer and mobile device. This means that we will need to add containers for different OS
- Users should also be able to make reservations for driving lessons as well as identify the driver the customer is scheduled to go with.
- The reservations are organized in three packages to which the user likes, depending on the user's availability.

## Requirements (Partially worked on it for future assignments)

### Nonfunctional Requirements

*In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.*

**Performance Requirements**

*What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?*

- The system needs to run off the web preferably over the cloud, according to Ian, the IT officer.
- Due to the DMV requirements, changes must be current with the DMV, so the system should run fast as well as update automatically with the DMV system.
- Notifications will also be sent whenever there is an update

**Platform Constraints**

*What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?*

- Driver-Pass should be able to access through any computer or mobile device. No security or backups will be needed as they do not want to deal with it.
- This also means containers will be utilized for computers and mobile OS.
- It also means that data will behave differently depending on the OS used to run Driver-Pass.

**Accuracy and Precision**

*How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?*

- To distinguish users, users will need to create an account to make reservations.
- Whenever a user wants to register, a phone call will be made out to the secretary and will have to show information like their full name, phone number, email, address, billing info etc.

**Adaptability**

*Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?*

- Yes, changes to the user will be allowed by letting Ian have full access to accounts (due to his IT officer role) so that whenever someone forgets their password or removing someone is needed, he is able to.
- The system should also be adaptable by having it connected to the DMV, so any new rules or regulations should be up to date.

**Security**

*What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?*

- Authentication and authorization will be used since Liam the owner, the IT officer, different employees, and users will all have different roles and accesses. They can utilize the least privilege rule to ensure no user has unauthorized access.
- For normal users, they will need their full name, address phone number, state, ID, credit card etc. In the case where the user forgets their login information, they will need to be able to automatically reset it (authentication as well).
- Ian, the IT officer, should be able to know how to make reservations, who canceled it or who modified theirs. Tracking allows for maintainability of the flow of the Driver-Pass website. The secretary will have access to phone calls and appointments.
- Or if they need to let go of someone (either a customer or an employee), Ian needs the ability to block their access. Security features like tracking should be included so that data management can be controlled and monitored.

**Functional Requirements**

*Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."*

- The system shall allow users to make, modify and cancel appointments and reservations
- The system shall allow customization of the packages like removals, creating new ones or changing them (in the future)
- The system shall allow users to update their password in case they forget their old one
- The system shall allow connectivity the the DMV's system to be up to date

**User Interface**

*What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?*

- The system will be hosted remotely, so the design must be responsive, real time syncing.
- There is an interface; it will contain the test progress of the user, tests the user has already completed, the time taken, the score and status (the status can be "not taken", "in progress", "failed", and "passed").
- The interface will also include the user's information, driver and student photos, notes (which contain comments by the drivers) and special needs.
- Other pages like an input form (used by the users and the secretary), a contact-us page, and a contact student page.

**Assumptions**

*What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?*

- One assumption is that the customer (Driver-Pass) won't ask for modifications after designing the class diagrams
- The system has to be flexible in order to customize packages, remove, or add new ones.
- Another assumption is the time frame of how long each design will take to create

## Limitations

*Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?*

- The system must run off the web (cloud-based system), and there is no focus on security and backup. This means there will be no databases, which means storage is very limited.
- No backups mean no database, and no database means there is a limit of how much data can enter the server at once since it will be hosted remotely.
- Requires a developer or a system analyst to add or remove modules for Driver-Pass
- Users won't be able to change or update their data if they are offline, which prevents data duplication.
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## Gantt Chart

